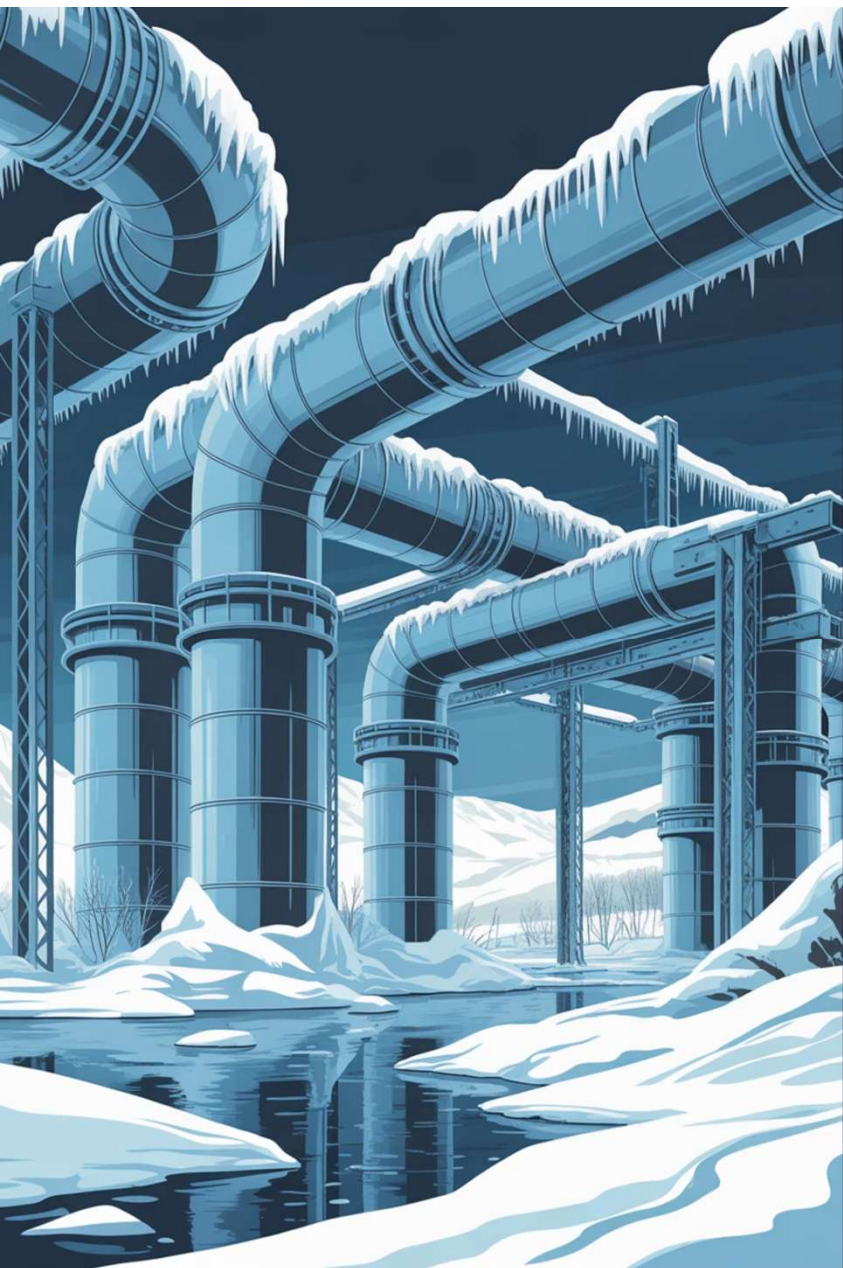


HEATSPOT OFF-GRID

Critical Off-Grid Thermal Protection

PIC-G21 · ElectroCap – 2026





CHAPTER 1.1 THE PROBLEM

Critical Infrastructure at Risk

Critical infrastructures in remote cold environments fail due to temperature.



Frozen Pipes
Ruptured tanks and burst pipes



Equipment Enclosure Freezing
Electronics and batteries lose performance



Telecom Outages
Remote connectivity becomes unreliable



Costly Repairs
Emergency maintenance expenses

There is no simple, autonomous, low-power spot heating solution for off-grid assets.



CHAPTER 1.2 THE PROBLEM

Agricultural Yield at Risk

Cold snaps and unstable temperatures threaten greenhouse yields and beehive productivity.



Frost Damage

Night-time temperature drops can permanently reduce yield and quality



Energy Limitations

Grid power is unavailable or too expensive and requires fixed infrastructure



Localised Needs

Only critical zones need protection



Beekeeping Impact

Cold nights reduce colony performance and increase winter losses

Agriculture and beekeeping need autonomous, low-power spot heating for critical protection off-grid.



CHAPTER 1.3 THE PROBLEM

BESS Performance at Risk

Cold temperatures reduce battery performance and can trigger failures in off-grid energy storage systems.



Capacity Drop

Reduced autonomy and unexpected shutdowns when energy is most needed.



Power Delivery Limits

Higher internal resistance impacts inverter performance and load support.



Protection Events and Downtime

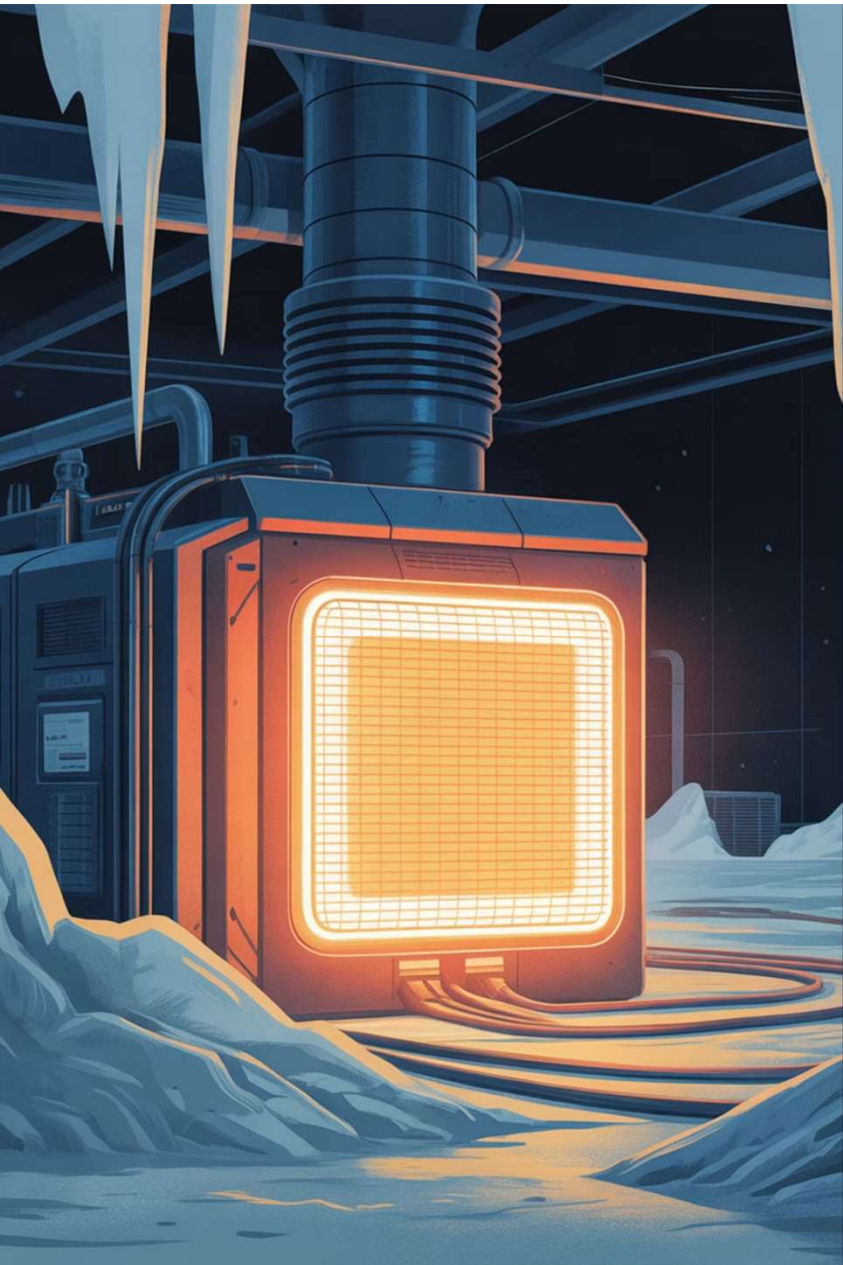
BMS may restrict operation at low temperature.



Costly Field Maintenance

Downtime, travel costs, and component stress increase total cost of ownership.

Off-grid BESS need targeted thermal protection to perform consistently, fully independent of the grid.



CHAPTER 4 VALUE PROPOSITION

Why We Are Different



Ultra Low Power
30-50W consumption



Localized Heating
Spot heating, not full enclosure



Multi-Day Autonomy
Solar-powered resilience

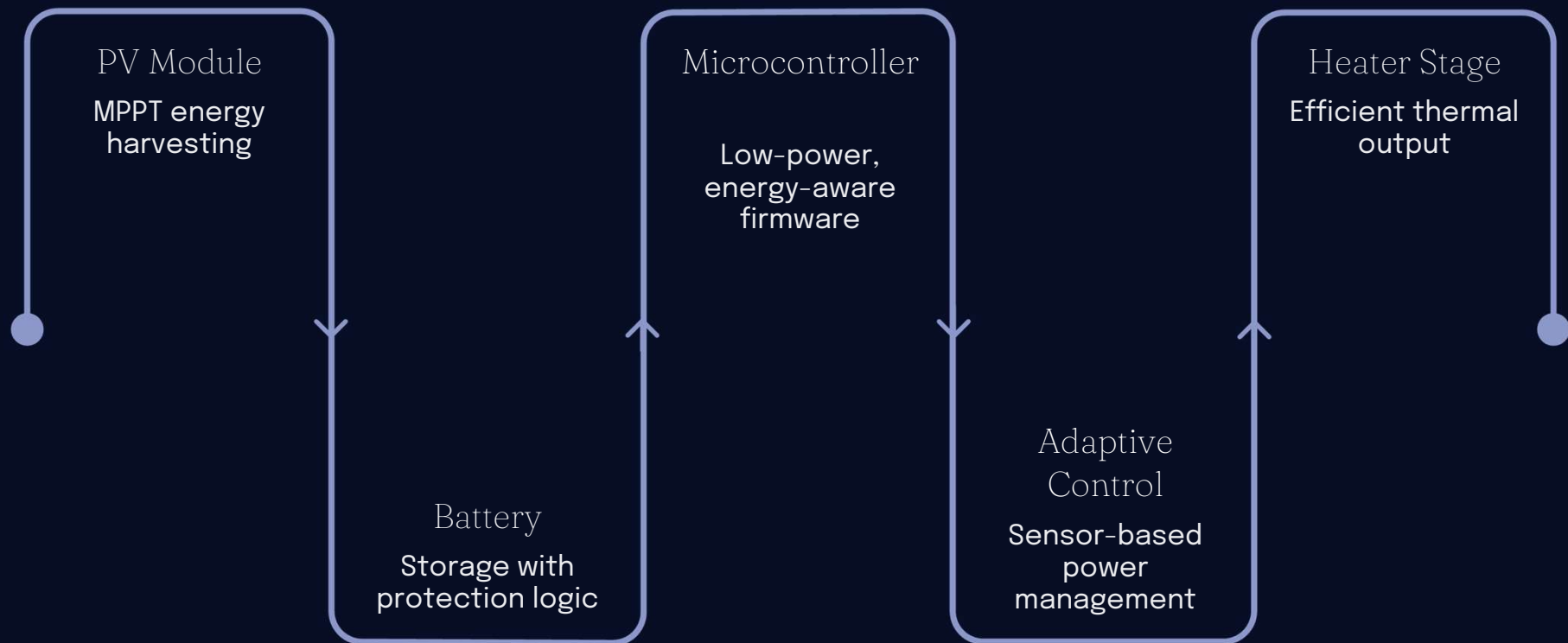


Modular & Scalable
Low install & maintenance cost

Resilience without grid dependency.

Technology Architecture

Integrated system with energy-aware firmware for maximum autonomy.



Each module communicates through an energy-aware firmware layer that dynamically balances harvesting, storage, and consumption.

Competitive Landscape

Alternative	Autonomous	Intelligent
Manual (blankets, gas)	✗	✗
Grid-connected industrial	✗	Partial
DIY battery heaters	Partial	✗
Niche solar beehive systems	✓	✗
PIC-G21 (Ours)	✓	✓

📄 **Innovation gap:** No modular, intelligent, low-power, asset-focused solution exists on the market today.

Technical Challenges

PV & Battery Sizing

Worst-case winter
scenarios

Low-Temp Battery
Mgmt

Safe operation below
0°C

Heater Drive
Electronics

Efficient power
delivery

Thermal Control
Algorithms

Stable, adaptive
regulation

Ultra-Low-Power
Firmware

Maximized energy
autonomy

We address these with **integrated energy modelling** and **robust embedded control**.

Validation Metrics

Data-driven validation, not qualitative claims.



Multi-Day Autonomy

Operation without sunlight



Temp Accuracy

*Maintain target temperature
within ± 2 °C*



Energy Ratio

Harvesting vs. consumption
balance



Fault Tolerance

Safe-state transitions



Comms Reliability

If remote monitoring enabled

Status of solution development

The project is currently in an **advanced development stage**, with its core technical and strategic foundations already established. The project is now moving into **optimisation and real-world validation**.



Functional Prototype

Developed and enabling initial validation of localised heating and off-grid energy management concepts.



System Architecture

Fully defined, covering energy generation, storage, and thermal control modules.



Bill of Materials

Complete BOM prepared, ensuring implementation feasibility and cost awareness.



Market Study

Conducted and confirming demand in agriculture, apiculture, and critical asset protection.

Deviations from the Initial Schedule

Weekly planning status against the initial project proposal (as of 7 Apr)

Period	Planned Scope	Status
P3-W1 to P3-W3	Literature review, scope refinement, revised proposal, website launch	4/4 completed
P3-W4	Stakeholder interviews planned, conducted, and summarized	3/3 completed
P3-W5	Requirements defined; use cases frozen	2/2 completed
P3-W6	BOM, prototype cost estimate, procurement requests initiated	3/3 completed
P3-W7	Mid-program pitch drafted; optional advisor draft not submitted	1/2* completed
P3-Prep	Final mid-program pitch finalized and submitted in Fenix	2/2 completed

* Planned as an optional advisor feedback step; not a mandatory deliverable and with no impact on the final submission.

Achieved results

A summary of results from the low-power, off-grid thermal protection system project – validating feasibility, market alignment, and the path forward.

System Feasibility

Successfully demonstrated a **low-power, off-grid thermal protection system** for critical applications, confirming the core concept is viable.

Controlled Heating

Initial tests validated the ability to deliver **controlled, localised heating** within the target power range of **30–50W**.

Integrated Prototype

Confirmed effective integration of **PV generation, battery storage, and intelligent control**, enabling energy-efficient operation.

Stable Temperature

Preliminary results show the system can **maintain stable temperature levels** under limited energy conditions.

Multi-Day Autonomy

The solution shows potential for **multi-day autonomy**, depending on environmental conditions.

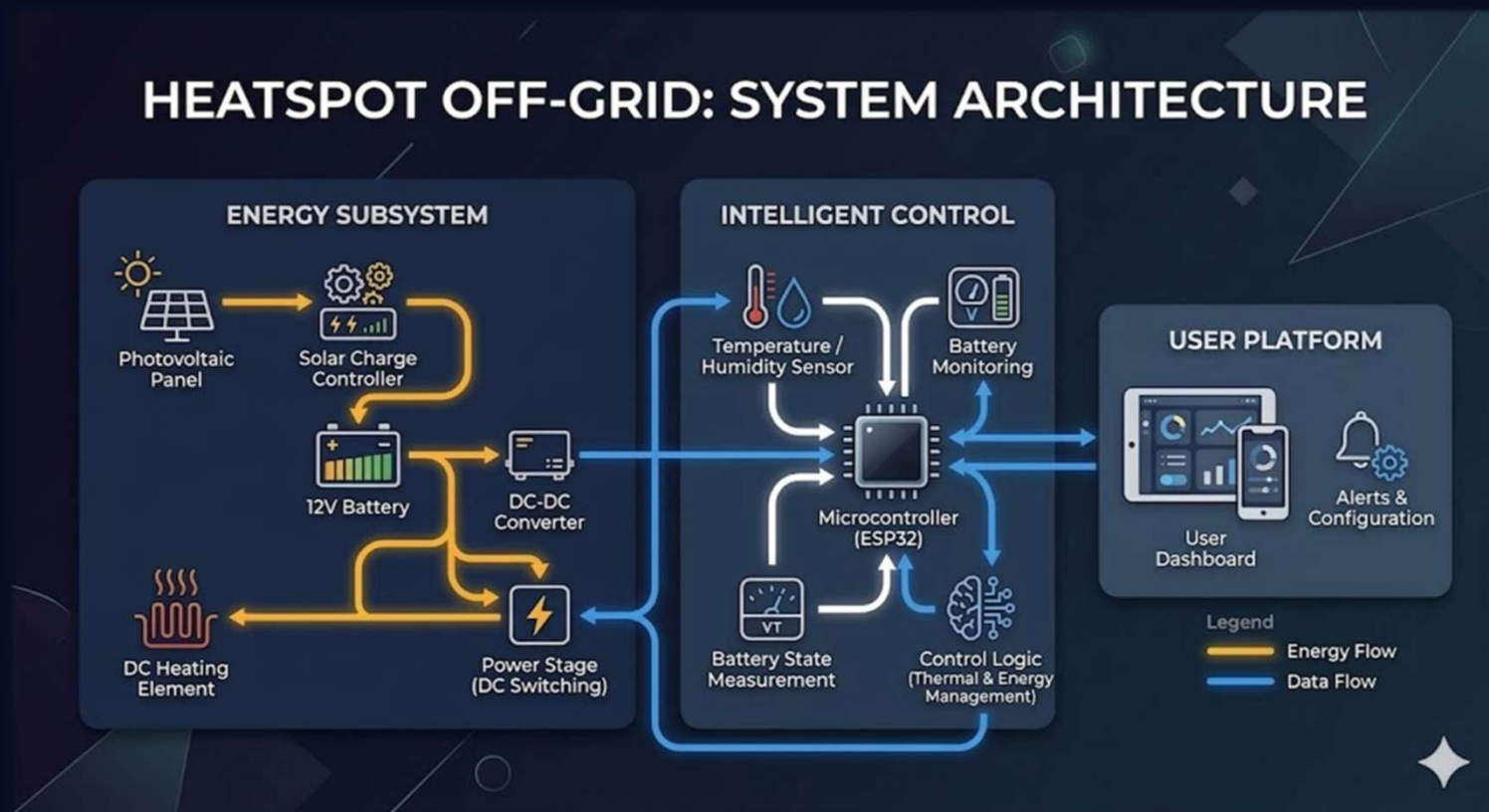
Market Focus

Engagement with **multiple companies and industry stakeholders** identified the **agriculture sector in Portugal** as the primary market opportunity.

❏ These results support the transition to field testing and system optimisation.

Diagram of the proposed solution architecture

Integrated system architecture combining energy, control and monitoring subsystems for autonomous thermal protection.



Technical and non-technical Challenges

Battery Selection
Challenge

Balancing weigh, cost and sub-zero performance

Solar controller: MPPT vs
PWM

Balancing conversion efficiency
vs system integration and cost

Solar sizing trade-off

Reduced panel to 30W due
to availability, requiring
requiring energy autonomy
targets

Market & Climate scope

Limited utility for Portugal's
mild climate except for
greenhouse agriculture

Modbus data gap

Sacrificed real-time Modbus
data for a zero-cost PWM,
impacting prototype ener

Contributions of each team member to the results

Work completed so far by team area

1

Catarina

Market Validation & External Engagement

- External Outreach: Engagement across relevant industry sectors to identify opportunities and build connections.
- Advisor Coordination: Coordinating with advisors and sharing insights with the broader team.
- Portugal Applications: Identification of high-potential applications and use cases within the Portuguese market.

2

Nicollas · Sébastien · Leonel

System Architecture & Technical Definition

- Requirements & Structuring: Defining project requirements and establishing a clear structural foundation for the system.
- Architecture & Use Cases: Consolidating system architecture and refining use cases for real-world applicability.
- Validation Metrics: Developing validation metrics and building the technical rationale underpinning the project.

3

Nicollas · Vasco

Prototype Planning, BOM & Resources

- BOM Definition: Defining the Bill of Materials and selecting the most suitable components for the prototype.
- Trade-off Analysis: Conducting technical and cost trade-off analysis to ensure optimal design decisions.
- Resource Optimisation: Maximising efficiency by leveraging existing lab equipment to reduce costs and accelerate development.

4

Sébastien · Leonel

Communication & Digital Support

- Website & IoT Interface: Development of the project website and early-stage IoT interface to support digital presence and functionality.
- Presentation Design: Creation of communication materials and presentation assets to convey the project clearly and professionally.
- Intermediate Delivery: Providing support for intermediate project milestones and ensuring timely, polished deliverables.

Contributions of each team member to the results

Balanced work allocation for prototype completion, validation and final presentation



Schedule of activities until the end of the project

Structured development roadmap with an **iterative build-test-improve cycle**.

